

Experiment 01: Overview of IoT Architecture & Protocols

1. 5-Tier Hierarchical IoT Architecture Reference Framework

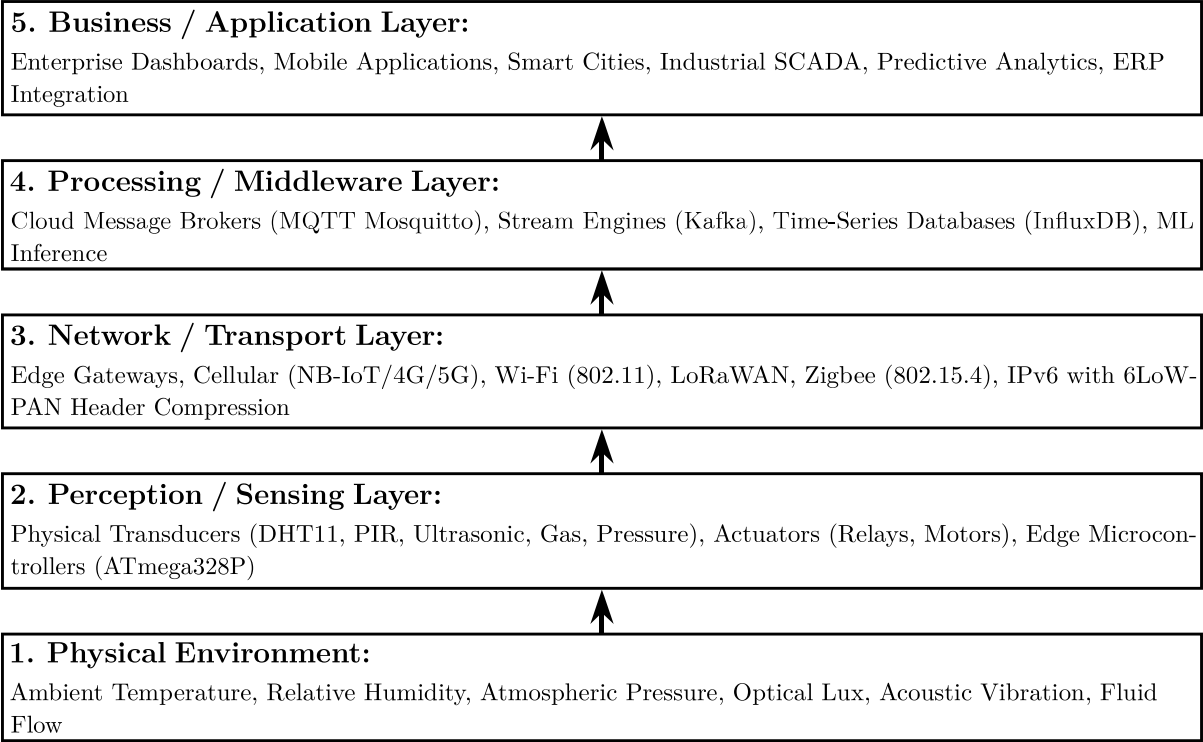


Figure 1.1: 5-Tier Hierarchical IoT Architecture Reference Framework.

2. Constrained IoT Communication Protocols Comparison

Protocol	Transport	Messaging Model	Header Size	Primary Application Domain
MQTT	TCP (1883)	Publish / Subscribe	2 Bytes (Fixed)	Real-time sensor telemetry, SCADA, battery-backed nodes.
CoAP	UDP (5683)	REST Request / Response	4 Bytes (Fixed)	Constrained 8-bit MCUs, low-power lossy networks (LLNs).
HTTP/REST	TCP (80/443)	Client / Server Request	> 500 B (Text)	Cloud-to-cloud APIs, enterprise REST integrations.
WebSocket	TCP (80)	Full-Duplex Stream	2–10 Bytes	Live browser telemetry monitoring dashboards.

Table 1.1: Comparison of Constrained Application Protocols in IoT Systems.

Experiment 02: Firmware Code, Waveforms & Truth Table

1. Complete Arduino C++ Firmware Code

Program Listing 2.1: Complete Arduino C++ Program for Dual Alternating LED Blinking

```
1 /*
2  * Experiment 02: Dual Alternating LED Interfacing with Arduino UNO
3  * Hardware Pin Connections: Pin D8 -> Green LED | Pin D7 -> Red LED
4  * Switching Interval: 1000 ms (1.0 second) Anti-Phase Blinking
5  */
6
7 const int LED1_PIN = 8;    // Green LED Channel (Digital Pin 8)
8 const int LED2_PIN = 7;    // Red LED Channel (Digital Pin 7)
9 const unsigned long BLINK_INTERVAL = 1000; // Period: 1000 ms (1.0 s)
10
11 void setup() {
12     // Configure both GPIO pins as digital output drivers
13     pinMode(LED1_PIN, OUTPUT);
14     pinMode(LED2_PIN, OUTPUT);
15
16     // Set deterministic initial state (both OFF)
17     digitalWrite(LED1_PIN, LOW);
18     digitalWrite(LED2_PIN, LOW);
19 }
20
21 void loop() {
22     // Phase 1: LED1 ON (Active HIGH), LED2 OFF (Active LOW)
23     digitalWrite(LED1_PIN, HIGH);
24     digitalWrite(LED2_PIN, LOW);
25     delay(BLINK_INTERVAL);
26
27     // Phase 2: LED1 OFF (Active LOW), LED2 ON (Active HIGH)
28     digitalWrite(LED1_PIN, LOW);
29     digitalWrite(LED2_PIN, HIGH);
30     delay(BLINK_INTERVAL);
31 }
```

2. Logic State Transitions & Timing Waveforms

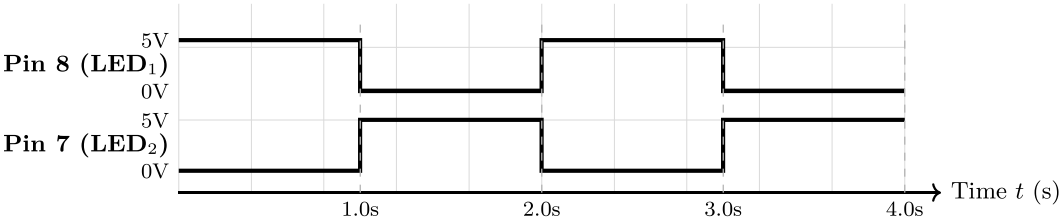


Figure 2.4: Digital Logic Timing Waveforms for Anti-Phase Switching ($T = 1.0\text{ s}$).

3. Circuit Logic Truth Table

Time Interval	Pin 8 Level	Pin 7 Level	LED ₁ State	LED ₂ State	Circuit Current & Physical Observation
0.0 s – 1.0 s	HIGH (5.0 V)	LOW (0.0 V)	ON	OFF	$I_{F1} \approx 3.0\text{ mA}$; Green light emission active.
1.0 s – 2.0 s	LOW (0.0 V)	HIGH (5.0 V)	OFF	ON	$I_{F2} \approx 3.0\text{ mA}$; Red light emission active.
2.0 s – 3.0 s	HIGH (5.0 V)	LOW (0.0 V)	ON	OFF	State repeats in strict periodic anti-phase.

Table 2.2: Truth Table and Circuit State Sequence.

Experiment 03: Complete Arduino C++ Source Code

Program Listing 3.1: Complete Arduino C++ Program for DHT11 Telemetry and Closed-Loop Alert Actuation

```
1  /*
2   * Experiment 03: DHT11 Sensor Interfacing with Closed-Loop High-Temp Alert
3   * Pin Connections: DHT11 DATA -> Pin 7 | Alert Red LED -> Pin 8
4   * Alarm Trigger Threshold: 30.0 deg C (Adjustable setpoint)
5   */
6
7  #include <DHT.h>
8
9  #define DHTPIN 7           // DHT11 DATA connected to Digital Pin 7
10 #define DHTTYPE DHT11     // Sensor model designation
11 #define ALERT_PIN 8       // Red Alert LED connected to Digital Pin 8
12 #define TEMP_THRESHOLD 30.0 // Closed-loop alarm trigger setpoint in Celsius
13
14 DHT dht(DHTPIN, DHTTYPE);
15
16 void setup() {
17     Serial.begin(9600);
18     pinMode(ALERT_PIN, OUTPUT);
19     digitalWrite(ALERT_PIN, LOW); // Initialize alert LED to OFF
20     dht.begin();
21
22     Serial.println("=====");
23     Serial.println("    IoT Lab: DHT11 Telemetry & Alert System    ");
24     Serial.println("=====");
25     Serial.println("Time(s)\tHumidity(%) \tTemp(C)\tAlert Status");
26 }
27
28 void loop() {
29     // DHT11 sampling rate is limited to 1 Hz; enforce 2.0 s polling interval
30     delay(2000);
31
32     float humidity = dht.readHumidity();
33     float temperature = dht.readTemperature();
34
35     // Validate sensor readout
36     if (isnan(humidity) || isnan(temperature)) {
37         Serial.println("ERROR: Failed to read from DHT11 sensor!");
38         return;
39     }
40
41     // Closed-Loop Threshold Comparison Logic
42     bool isAlert = (temperature >= TEMP_THRESHOLD);
43     digitalWrite(ALERT_PIN, isAlert ? HIGH : LOW);
44
45     // Stream formatted telemetry to Serial Monitor
46     Serial.print(millis() / 1000);
47     Serial.print(" s\t");
48     Serial.print(humidity, 1);
49     Serial.print(" %\t\t");
50     Serial.print(temperature, 1);
51     Serial.print(" C\t");
52     Serial.println(isAlert ? "ACTIVE [ALERT ON]" : "NORMAL [OFF]");
53 }
```

Experiment 03: Telemetry Observations & Serial Output

1. Experimental Telemetry Observations & Checksum Verification

S.No	Thermal State	Temp	Humidity	Alert Status ($T \geq 30^{\circ}\text{C}$)	Checksum Parity Calculation
1	Ambient Lab Environment	27.4°C	62.0%	OFF (Normal)	$0x1B + 0x00 + 0x3E + 0x00 = 0x59$ [PASS]
2	Ambient Lab Environment	28.1°C	60.5%	OFF (Normal)	$0x1C + 0x00 + 0x3C + 0x00 = 0x58$ [PASS]
3	Elevated Heat Applied	31.8°C	74.0%	ON (ACTIVE ALERT)	$0x1F + 0x00 + 0x4A + 0x00 = 0x69$ [PASS]
4	Peak Heated State	34.0°C	86.0%	ON (ACTIVE ALERT)	$0x22 + 0x00 + 0x56 + 0x00 = 0x78$ [PASS]
5	Post-Cooling Normal	28.5°C	65.0%	OFF (Normal)	$0x1C + 0x00 + 0x41 + 0x00 = 0x5D$ [PASS]

Table 3.2: Measured Environmental Telemetry and Closed-Loop Alert Evaluation Log.

2. Raw Serial Monitor Terminal Output (9600 Baud)

Serial Monitor Terminal Capture: Raw Telemetry Stream (9600 Baud)

=====				
IoT Lab: DHT11 Telemetry & Alert System				
=====				
Time(s)	Humidity(%)	Temp(C)	Alert Status	
2 s	62.0 %	27.4 C	NORMAL [OFF]	
4 s	60.5 %	28.1 C	NORMAL [OFF]	
6 s	74.0 %	31.8 C	ACTIVE [ALERT ON]	<-- [Thermal Stimulus Applied]
8 s	86.0 %	34.0 C	ACTIVE [ALERT ON]	<-- [Peak Heating Condition]
10 s	65.0 %	28.5 C	NORMAL [OFF]	<-- [Normalized Below 30 C]